Generating random graphs with given properties

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The bounty expires tomorrow. Answers to this question are eligible for a +50 reputation bounty. David Scholz is looking for a **canonical answer**.

One way to define random graphs is to fix a finite set of properties P. Picking with equal probability a graph with $|V| = n$ out of the set $\Gamma_n(P)$ of all n-(vertex-)graphs with properties P yields a random n-P-graph. This is the theoretical point of view. Practically, the set is not at hand and one has to *generate* with equal probability graphs that happen to be in $\Gamma_n(P)$.

The classical Erdös-Renyi graphs $G(n, m)$ are defined like this with $P = \{|E| = m\}$. To generate an Erdös-Renyi random graph is easy.

Configuration models allow to generate random graphs with a given degree distribution. Advanced configuration models allow to generate graphs that in addition have a given distribution of the clustering coefficient.

My questions are:

- For which classes of random graphs (defined by a set of properties they must have) are there well known and efficient ways to generate them randomly? (This might be a big list question.)
- For which classes of random graphs are **no** ways known to efficiently generate them randomly?
- By which heuristic arguments can one tell whether for a given set of (non-contradictory) properties there may be ways to efficiently generate them randomly - or not?

graph-theory big-list random-graphs

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Hans-Peter Stricker
16.2k 4 51 109

2 Small bit of pedantry: presumably you want your set $\Gamma(P)$ to be finite and every graph in it to be generated with equal probability? Things get Complicated if $\Gamma(P)$ is allowed to be infinite... – Steven Stadnicki Mar 17 at 16:18

2 @StevenStadnicki: Thanks, I made these assumptions explicit. – Hans-Peter Stricker Mar 17 at 16:21

1 You might still want to clarify a little bit — P can be finite without $\Gamma(P)$ being finite. (e.g., if P is 'G is cubic'). Does it make sense in your context to specifically consider the set $\Gamma_n(P)$ of n-vertex graphs with the property P? (This would make the Erdös-Renyi graphs $\Gamma_n(P)$ with $P = \{|E| = m\}$, for instance.) – Steven Stadnicki Mar 17 at 16:34

@StevenStadnicki: Again you are right! I'll correct this later. – Hans-Peter Stricker Mar 17 at 16:58

2 Another pragmatic facet of this is whether you want distributions that are exactly random or merely random

within a certain tolerance. I'm having a hard time finding good references but I've seen Metropolis-Hastings sampling used for sampling constrained configurations (e.g. colorings of a graph) where there are transforms that can take one valid configuration to another. See for instance chapter 2 of uni-ulm.de/fileadmin/website_uni_ulm/mawi.inst.110/lehre/ss15/... . – Steven Stadnicki Mar 17 at 17:04

@StevenStadnicki: I've done so. – Hans-Peter Stricker Mar 18 at 10:47